BFS-U3-04S2C-C USB 3.1, USB 3.1 Blackfly® S, Color Camera

#22-085



Teledyne FLIR IIS Blackfly® S USB3 Camera (front)

Phone: 1-800-363-1992 or 1-856-573-6250, Fax: 1-856-573-6295
Edmund Optics India Private Limited, #267, Greystone Building, Second Floor, 6th Cross Rd, Binnamangala, Stage 1, Indiranagar, Bengaluru, Karnataka, India 560038
www.edmundoptics.in

Teledyne FLIR IIS Blackfly S: Advanced Machine Vision Cameras with powerful features

Capture the images you need from advanced sensors in enclosed or board-level configurations

The **Blackfly® S** is a versatile and compact machine vision camera series that leverages the industry's most advanced area scan sensors in an ultra-compact form factor. It combines powerful features that easily produce the exact images required, accelerating application development. Combining both automatic and precise manual controls over image capture and on-camera pre-processing. With options ranging from high-speed performance, high-resolution images, polarization, or low-light sensitivity, the Blackfly® S series of cameras can deliver the required results.

With the selection of camera variations all sharing the same form factor, it makes it easy to develop once, deploy anywhere. On camera features include IEEE1588 clock synchronization and full compatibility with popular third-party software supporting either GigE Vision or USB3 Vision interfaces. The Blackfly® S is available in GigE, USB3, cased, and board-level versions.

Note: USB3 cable sold separately. Use (#03-618) 5mm Spacer to convert CS-Mount Cameras to C-Mount. Software available for download.

U.S. Tariff Pricing Notice: Pricing for certain products in this family may reflect recent changes in U.S. tariffs on Canadian-origin products. **Prices shown include any applicable tariff-related adjustment; no additional tariff surcharge will be added at checkout.**

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Spectrum:	Color

General

Type:	Color Camera
Model Number:	BFS-U3-04S2C-C
Manufacturer:	FLIR
Camera Series:	Blackfly® S

Physical & Mechanical Properties

Dimensions (mm):	29 x 29 x 30 (excludes connectors and lens mount)
Weight (g):	36
Housing:	Full

Sensor

Image Buffer:	240MB
Sensor Format:	1/2.9"
Resolution (Megapixels):	0.40
Frame Rate (fps):	522.00
Pixels (H x V):	720 x 540
Pixel Size, H x V (µm):	6.9 x 6.9
Sensing Area, H x V (mm):	4.97 x 3.73
Imaging Sensor:	Sony IMX287
Type of Sensor:	Progressive Scan CMOS
Shutter Type:	Global
Pixel Depth:	8/10/12 Bit
Exposure Time:	4µs - 30s
Dynamic Range (dB):	74.35
Machine Vision Standard:	USB3 Vision v1.0

Electrical

Power Consumption (W):	3

Hardware & Interface Connectivity

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Interface:	USB 3.1 Gen 1
Connector:	USB 3.1 Gen 1, Micro-B with Screw Locks
Power Supply:	Power over USB or via GPIO with #88-063
GPIOs:	1 opto-isolated input, 1 opto-isolated output, 1 non-isolated bi-directional, 1 non-isolated input
Synchronization:	Hardware Trigger (GPIO) or Software Trigger
Interface Port Orientation:	Back Panel
GPIO Connector Type:	6-pin Hirose (HR10)

Threading & Mounting

Mount:	C-Mount
Mounting Threads:	¼-20 with Tripod Mount Adapter #88-210

Environmental & Durability Factors

Operating Temperature (°C):	0 to +50
Storage Temperature (°C):	-30 to +60

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September 03, 2026

To Whom It May Concern,

This document certifies that the product stated below has been reviewed as requested by Edmund Optics:

Stock No.	*Description*
22-085	BFS-U3-04S2C-C USB 3.1, USB 3.1 Blackfly® S, Color Camera

This item is EU RoHS (2015/863/EU) exempt based on **Exemption 6c**: Copper alloy containing up to 4% lead by weight, **Exemption 7a**: lead in high melting temperature type solders (i.e. lead- based alloys containing 85 % by weight or more lead), and **Exemption 7c-I**: electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.



Jay Budd, Director of Corporate Compliance
September 03, 2026

Edmund Optics Inc. - 101 E Gloucester Pike, Barrington, NJ 08007 | 1-800-363-1992 | Compliance@edmundoptics.com

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Edmund Optics certifies that all articles included in this shipment are in compliance with the terms and conditions of this order. The company also certifies that the articles included in this shipment are in accordance to all agreed upon specifications and quality assurance provisions. Please call 1-800-363-1992 should any questions arise from this shipment.



Jeff Harvey- EVP of Operations

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This item contains substances listed on the REACH SVHC list as of 23 January 2024 in quantities exceeding 0.1% weight by weight, please contact compliance@edmundoptics.com if you require more information.

Please note the following:

- Entry 11 of Annex V of the REACH regulation EC 1907/2006 as amended by regulation 987/2008, exempts “non-hazardous” glass components from registration.
- Items considered to be packing materials and which do not form part of, or are not required for the operation of, the product, are not considered to be within scope of this declaration.
- EO does not routinely analyze products for substances not purposely added or specified within the design specifications, drawings and assembly procedures.

This certification means that:

- EO's suppliers have confirmed the material composition of this product.
- EO has implemented rigorous procedures to document this compliance.
- The information provided may, or may not, be based upon actual test data, or on information from our Vendors, Raw Material Suppliers or Subcontractors.

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